

1. Connected two point correlation functions

First we choose interpolating operators with the quantum numbers of the meson we would like to study:

$$O_M = \bar{\psi}^{(1)}(x)\Gamma\psi^{(2)}(x) \quad (1)$$

$$\bar{O}_M = \bar{\psi}^{(2)}(x)\bar{\Gamma}\psi^{(1)}(x) \quad (2)$$

where $\bar{\Gamma} = \gamma_0\Gamma^\dagger\gamma_0$. The gamma matrix is chosen to have the same J^{PC} quantum numbers as the meson we want to create. We then calculate the expectation value to create a meson at y and destroy it again at x :

$$\langle O_M(x)\bar{O}'_M(y) \rangle = \langle \bar{\psi}^{(1)}(x)\Gamma\psi^{(2)}(x)\bar{\psi}^{(2)}(y)\bar{\Gamma}'\psi^{(1)}(y) \rangle \quad (3)$$

$$= \Gamma_{\alpha\beta}\bar{\Gamma}'_{\gamma\delta}\langle \bar{\psi}_\alpha^{(1)}(x)\psi_\beta^{(2)}(x)\bar{\psi}_\gamma^{(2)}(y)\psi_\delta^{(1)}(y) \rangle \quad (4)$$

$$= -\Gamma_{\alpha\beta}\bar{\Gamma}'_{\gamma\delta}\langle \psi_\beta^{(2)}(x)\bar{\psi}_\gamma^{(2)}(y)\psi_\delta^{(1)}(y)\bar{\psi}_\alpha^{(1)}(x) \rangle \quad (5)$$

where the sign in the last line comes from exchanging the anticommuting $\bar{\psi}_\alpha$ 3 times. Wick contracting where $\langle \psi_\alpha(x)\bar{\psi}_\beta(y) \rangle = S_{\alpha\beta}(x, y)$ gives

$$\langle O_M(x)\bar{O}'_M(y) \rangle = -\Gamma_{\alpha\beta}S_{\beta\gamma}^{(2)}(x, y)\bar{\Gamma}'_{\gamma\delta}S_{\delta\alpha}^{(1)}(y, x) \quad (6)$$

$$= -\text{Tr} \left[\Gamma S^{(2)}(x, y)\bar{\Gamma}' S^{(1)}(y, x) \right] \quad (7)$$

To get correlation functions we Fourier transform, ie. sum over all source and sink separations while projecting onto the momentum we want to give to the particle:

$$C(t - \tau, \vec{p}) = \sum_{\vec{x}\vec{y}} e^{-i\vec{p}(\vec{x}-\vec{y})} \langle O_M(\vec{x}, t) O'_M(\vec{y}, \tau) \rangle \quad (8)$$

$$= -\sum_{\vec{x}\vec{y}} e^{-i\vec{p}(\vec{x}-\vec{y})} \text{Tr} \left[\Gamma S^{(2)}(x, y)\bar{\Gamma}' S^{(1)}(y, x) \right] \quad (9)$$

here $x = (\vec{x}, t)$ and $y = (\vec{y}, \tau)$ and the zero momentum correlator is:

$$C(t - \tau, 0) = -\sum_{\vec{x}\vec{y}} \text{Tr} \left[\Gamma S^{(2)}(x, y)\bar{\Gamma}' S^{(1)}(y, x) \right] \quad (10)$$

We now use γ_5 Hermiticity: $\gamma_5 S^\dagger(x, y)\gamma_5 = S(y, x)$,

$$C(t - \tau, 0) = -\sum_{\vec{x}\vec{y}} \text{Tr} \left[\gamma_5 \Gamma S^{(2)}(x, y)\bar{\Gamma}' \gamma_5 S^{\dagger(1)}(x, y) \right] \quad (11)$$

2. Point Sources

Using a delta function source and solving the Dirac equation gives a point propagator,

$$D_{\alpha a, \beta b}(x, y) S_{\beta b, \gamma c}(y, z) = \delta(x, z) \delta_{ac} \delta_{\alpha\gamma} \quad (12)$$

usually $z = (\vec{0}, 0)$ so we get $S(y, 0) = \gamma_5 S^\dagger(0, y)\gamma_5$. Then we use these to calculate correlation functions,

$$C(t, 0) = -\sum_{\vec{x}} \text{Tr} e^{-i\vec{p}\vec{x}} \left[\gamma_5 \Gamma S(x, 0) \bar{\Gamma}' \gamma_5 S(x, 0) \right] \quad (13)$$

Translational invariance in the limit of infinitely many gauge configurations implies $S(x, y) = S(|x - y|)$, so the sum over $\vec{y}$ in equation (??) just gives V times equation ???. We place the source at the time origin so $\tau = 0$.

3. One-end Trick

For this method it helps to write all the indices out,

$$C(t - \tau, 0) = - \sum_{\vec{x}\vec{y}} (\gamma_5 \Gamma)_{\alpha\beta} S_{\beta\gamma,bc}^{(2)}(x, y) (\bar{\Gamma}' \gamma_5)_{\gamma\delta} S_{\delta\alpha,cb}^{\dagger(1)}(x, y) \quad (14)$$

Greek indices $\alpha, \beta, \gamma, \delta, \dots$ are spinor indices and Latin indices $a, b, c, d, \dots$ are colour indices. The one-end trick involves inserting a delta function in colour, spin and space.

$$C(t - \tau, 0) = - \sum_{\vec{x}\vec{y}\vec{z}} (\gamma_5 \Gamma)_{\alpha\beta} S_{\beta\gamma,bc}^{(2)}(\vec{x}, t; \vec{y}, \tau) \delta_{\gamma\lambda} \delta_{cd} \delta(\vec{y}, \vec{z}) (\bar{\Gamma}' \gamma_5)_{\lambda\delta} S_{\delta\alpha,db}^{\dagger(1)}(\vec{x}, t; \vec{z}, \tau) \quad (15)$$

The delta function is approximated with a $Z(2) \times Z(2)$ noise source on timeslice τ

$$\delta_{\gamma\lambda} \delta_{cd} \delta(\vec{y}, \vec{z}) \approx \frac{1}{K} \sum_{k=0}^K |\eta_{\gamma c}^{(k)}(\vec{y})\rangle \langle \eta_{\lambda d}^{(k)}(\vec{z})| \quad (16)$$

which is exact in the limit $K \rightarrow \infty$.

$$C(t - \tau, 0) = - \frac{1}{K} \sum_{k=0}^K \sum_{\vec{x}\vec{y}\vec{z}} (\gamma_5 \Gamma)_{\alpha\beta} S_{\beta\gamma,bc}^{(2)}(\vec{x}, t; \vec{y}, \tau) |\eta_{\gamma c}^{(k)}(\vec{y})\rangle \langle \eta_{\lambda d}^{(k)}(\vec{z})| (\bar{\Gamma}' \gamma_5)_{\lambda\delta} S_{\delta\alpha,db}^{\dagger(1)}(\vec{x}, t; \vec{z}, \tau) \quad (17)$$

Defining

$$\phi_{\beta,b}^{(k)}(\vec{x}, t; \tau) = \sum_{\vec{y}} S_{\beta\gamma,bc}^{(2)}(\vec{x}, t; \vec{y}, \tau) |\eta_{\gamma c}^{(k)}(\vec{y})\rangle \quad (18)$$

and

$$\phi_{\alpha,b}^{\Gamma(k)}(\vec{x}, t; \tau) = \sum_{\vec{z}} S_{\alpha\delta,bd}^{(1)}(\vec{x}, t; \vec{z}, \tau) (\bar{\Gamma}' \gamma_5)_{\delta\lambda}^\dagger |\eta_{\lambda d}^{(k)}(\vec{z})\rangle \quad (19)$$

the correlator can be evaluated as,

$$C(t - \tau, 0) = - \frac{1}{K} \sum_{k=0}^K \sum_{\vec{x}} (\gamma_5 \Gamma)_{\alpha\beta} \phi_{\beta,b}^{(k)}(\vec{x}, t; \tau) \phi_{\alpha,b}^{\Gamma(k)}(\vec{x}, t; \tau) \quad (20)$$

3.1. One-End-Trick in HiRep

In HiRep the code `Spectrum/mk_mesons_with_z2semwall.c` does two solves to calculate $S^{(1)}|\eta\rangle$ and $S^{(2)}(\bar{\Gamma}' \gamma_5)^\dagger |\eta\rangle$. HiRep has

$$\rho = \rho_c(\vec{y}) \quad (21)$$

a $Z(2) \times Z(2)$ colour vector at all (even) spatial sites $\vec{y}$ and non-zero only on timeslice τ .

$$\rho_\beta^\alpha = \delta_{\alpha\beta} \rho^\alpha \quad (22)$$

eg.

$$\rho_\beta^1 = \begin{pmatrix} 0 \\ \rho^1 \\ 0 \\ 0 \end{pmatrix}$$

It then solves for the four objects

$$\chi_\beta^\alpha = S_{\beta\gamma} \rho_\gamma^\alpha \quad (23)$$

eg.

$$\chi_\beta^0 = \begin{pmatrix} S_{00} \rho^0 \\ S_{10} \rho^0 \\ S_{20} \rho^0 \\ S_{30} \rho^0 \end{pmatrix}$$

For every different Γ that is required it does four more inversions,

$$\chi_\beta^{\Gamma\alpha} = S_{\beta\gamma}(\Gamma\gamma_5)^\dagger_{\gamma\delta}\rho_\delta^\alpha \quad (24)$$

before calculating the correlator as,

$$C(t - \tau, 0) = -\frac{1}{K} \sum_{k=0}^K \sum_{\lambda=0}^3 \sum_{\vec{x}} (\gamma_5 \Gamma)_{\alpha\beta} \chi_{\beta,b}^\lambda(\vec{x}, t; \tau) \chi_{\alpha,b}^{\Gamma\lambda\dagger}(\vec{x}, t; \tau) \quad (25)$$

where the λ sum is over the 4 spinor components.

We should be able to improve the signal and reduce the number of inversions with two modifications. First, instead of having a different noise vector for every spin component we reuse the same noise, i.e.

$$\rho_\beta^\alpha = \delta_{\alpha\beta} \rho \quad (26)$$

for fixed ρ . Using less noise seems to be generally preferred.

Secondly there is no need to invert for every different Γ . Let,

$$\chi_\beta^{\Gamma\alpha} = (\Gamma\gamma_5)^\dagger_{\gamma\alpha} \chi_\beta^\gamma \quad (27)$$

This is true because,

$$(\Gamma\gamma_5)^\dagger_{\gamma\alpha} \chi_\beta^\gamma = (\Gamma\gamma_5)^\dagger_{\gamma\alpha} S_{\beta\delta} \rho_\delta^\gamma = (\Gamma\gamma_5)^\dagger_{\gamma\alpha} S_{\beta\delta} \delta_{\gamma\delta} \rho = (\Gamma\gamma_5)^\dagger_{\gamma\alpha} S_{\beta\gamma} \rho = S_{\beta\gamma} (\Gamma\gamma_5)^\dagger_{\gamma\alpha} \rho \quad (28)$$

then the correlation function is

$$C(t - \tau, 0) = -\frac{1}{K} \sum_{k=0}^K \sum_{\lambda=0}^3 \sum_{\vec{x}} (\gamma_5 \Gamma)_{\alpha\beta} \chi_{\beta,b}^\lambda(\vec{x}, t; \tau) \chi_{\alpha,b}^{\Gamma\lambda\dagger}(\vec{x}, t; \tau) \quad (29)$$

as before. By using the `spin_matrix` object in HiRep to construct the objects $\chi_{\beta,b}^\lambda$ the correlators can be calculated with only $4N_F$ inversions.

4. Disconnected

The Disconnected contributions occur when we have fermion species of the same type in the hadron interpolator O_M :

$$O_M(x) = \bar{\psi}(x) \Gamma \psi(x) \quad (30)$$

The same manipulations that lead to equation (5) give,

$$\langle O_M(x) \bar{O}'_M(y) \rangle = \langle \bar{\psi}(x) \Gamma \psi(x) \bar{\psi}(y) \bar{\Gamma}' \psi(y) \rangle \quad (31)$$

$$= \Gamma_{\alpha\beta} \bar{\Gamma}'_{\gamma\delta} \langle \bar{\psi}_\alpha(x) \psi_\beta(y) \bar{\psi}_\gamma(y) \psi_\delta(x) \rangle \quad (32)$$

There are two allowed Wick contractions,

$$\langle O_M(x) \bar{O}'_M(y) \rangle = -\text{Tr} [\Gamma S(x, y) \bar{\Gamma}' S(y, x)] + \text{Tr} [\Gamma S(x, x)] \text{Tr} [\bar{\Gamma}' S(y, y)] \quad (33)$$

the connected and disconnected contributions. Fourier transforming the first term gives us the same result as before. For the disconnected part,

$$D(t - \tau, \vec{p}) = \sum_{\vec{x}\vec{y}} e^{-i\vec{p}(\vec{x}-\vec{y})} \text{Tr} [\Gamma S(x, x)] \text{Tr} [\bar{\Gamma}' S(y, y)] \quad (34)$$

again $x = (\vec{x}, t)$ and $y = (\vec{y}, \tau)$, the zero momentum correlator is

$$D(t - \tau, 0) = \sum_{\vec{x}\vec{y}} \text{Tr} [\Gamma S(x, x)] \text{Tr} [\bar{\Gamma}' S(y, y)] \quad (35)$$

$$= \sum_{\vec{x}} \text{Tr} [\Gamma S(x, x)] \sum_{\vec{y}} \text{Tr} [\bar{\Gamma}' S(y, y)] \quad (36)$$

This means we have to evaluate objects like,

$$d(t) = \sum_{\vec{x}} \text{Tr} [\Gamma S(x, x)] = \sum_{\vec{x}} \Gamma_{\alpha\beta} S_{\beta\alpha}(x, x) \quad (37)$$

Using

$$\phi_{\beta,b}^{(k)}(\vec{x}, t; \tau) = \sum_{\vec{y}} S_{\beta\gamma,bc}(\vec{x}, t; \vec{y}, \tau) |\eta_{\gamma c}^{(k)}(\vec{y})\rangle \quad (38)$$

which implies

$$\frac{1}{K} \sum_k \sum_{\vec{x}} \text{Tr} \left[\langle \eta^{(k)}(\vec{x}) | \Gamma | \phi^{(k)}(\vec{x}, t; \tau) \rangle \right] = \frac{1}{K} \sum_k \sum_{\vec{x}} \Gamma_{\beta\gamma} |\phi_{\gamma,b}^{(k)}(\vec{x}, t; \tau)\rangle \langle \eta_{\beta b}^{(k)}(\vec{x})| \quad (39)$$

$$= \frac{1}{K} \sum_k \sum_{\vec{x}} \Gamma_{\beta\gamma} \sum_{\vec{y}} S_{\gamma\alpha,bc}(\vec{x}, t; \vec{y}, \tau) |\eta_{\alpha c}^{(k)}(\vec{y})\rangle \langle \eta_{\beta b}^{(k)}(\vec{x})| \quad (40)$$

Using the limit $K \rightarrow \infty$ this becomes,

$$\sum_{\vec{x}} \Gamma_{\beta\gamma} \sum_{\vec{y}} S_{\gamma\alpha,bc}(\vec{x}, t; \vec{y}, \tau) \delta_{bc} \delta_{\alpha\beta} \delta(\vec{y}, \vec{x}) = \sum_{\vec{x}} \Gamma_{\beta\gamma} S_{\gamma\beta,bb}(\vec{x}, t; \vec{x}, \tau) \quad (41)$$

$$d(t, \tau) = \text{Tr} [\Gamma S(\vec{x}, t; \vec{x}, \tau)] \quad (42)$$

We want only cases where $t = \tau$ so we need either (a) four noise vectors on every timeslice or (b) noise vectors that are nonzero on all timeslices. In case (b) we would evaluate,

$$\frac{1}{K} \sum_k \sum_{\vec{x}} \text{Tr} \left[\langle \eta^{(k)}(\vec{x}, t) | \Gamma | \phi^{(k)}(\vec{x}, t) \rangle \right] \quad (43)$$

with

$$\phi_{\beta,b}^{(k)}(\vec{x}, t) = \sum_{\vec{y}} S_{\beta\gamma,bc}(\vec{x}, t; \vec{y}, \tau) |\eta_{\gamma c}^{(k)}(\vec{y}, \tau)\rangle \quad (44)$$

5. Cancelling Backwards Propagation

The two point function evaluated in the centre of the lattice is (including the backward propagating part to give the extra factor of 2),

$$C(T/2, \vec{p}) = \frac{|Z_\pi|^2}{2E_\pi(\vec{p})} 2e^{-E_\pi(\vec{p})(T/2)} \quad (45)$$

therefore

$$\frac{1}{2} C(T/2, \vec{p}) e^{-E_\pi(\vec{p})(T/2-t)} = \frac{|Z_\pi|^2}{2E_\pi(\vec{p})} e^{-E_\pi(\vec{p})(T-t)} \quad (46)$$

then

$$C_{\rightarrow}(t, \vec{p}) = C(t, \vec{p}) - \frac{1}{2} C(T/2, \vec{p}) e^{-E_\pi(\vec{p})(T/2-t)} \quad (47)$$

is the forward propagating part only. $E_\pi(\vec{p})$ is obtained by fitting the zero momentum correlator and using $E(\vec{p}) = \sqrt{m_\pi^2 + \vec{p}^2}$. The factor $C(T/2, \vec{p})$ can be obtained also from the zero momentum correlator, by fitting to obtain $|Z_\pi|^2$ and using the fact that this is momentum independent. Since 0 momentum results are used this might not be too noisy.

Alternatively the Wilson action is invariant under

$$\psi(x) \rightarrow \mathcal{P}_\mu[\psi(x)] = \gamma_\mu \psi(P_\mu[x]) \quad (48)$$

$$\bar{\psi}(x) \rightarrow \mathcal{P}_\mu[\bar{\psi}(x)] = \bar{\psi}(P_\mu[x]) \gamma_\mu \quad (49)$$

where $P_\mu[x]$ reverses the sign of all the components of x except the μ one. Time reversal corresponds to $\mathcal{T} = \mathcal{P}_1\mathcal{P}_2\mathcal{P}_3$.

$$\psi(x) \rightarrow \mathcal{T}[\psi(x)] = \gamma_0\gamma_5\psi(T[x]) \quad (50)$$

$$\bar{\psi}(x) \rightarrow \mathcal{T}[\bar{\psi}(x)] = \bar{\psi}(T[x])\gamma_5\gamma_0 \quad (51)$$

Using this the T symmetry of operators used to construct the correlators can be calculated to calculate the sign on the backwards propagating part.

$$\langle O_1(t)O_2(0) \rangle = C(t) = A \left(e^{-Et} + \tau_1\tau_2 e^{-E(T-t)} \right) \quad (52)$$

where $\tau_i = \pm 1$ is the $\mathcal{T}$ eigenvalue of O_i .

We mostly use correlators where $O_1 = O_2$ so $\tau_1\tau_2 = 1$ then the correlator is

$$C_{pp}(t, \vec{p}) = \frac{|Z_\pi|^2}{2E_\pi(\vec{p})} \left(e^{-E_\pi(\vec{p})t} + e^{-E_\pi(\vec{p})(T-t)} + e^{-E_\pi(\vec{p})(2T-t)} + \dots \right) \quad (53)$$

The subscript on C_{pp} refers to the fact that both propagators used periodic boundary conditions. We want to cancel the backwards propagating part which can be done by solving the forward propagator $S(0, x)$ using antiperiodic time bc's and the backward $S(x, 0)$ with periodic time bc's to give an extra minus sign,

$$C_{ap}(t, \vec{p}) = \frac{|Z_\pi|^2}{2E_\pi(\vec{p})} \left(e^{-E_\pi(\vec{p})t} - e^{-E_\pi(\vec{p})(T-t)} + e^{-E_\pi(\vec{p})(2T-t)} - \dots \right) \quad (54)$$

so,

$$C_{ap}(t, \vec{p}) + C_p(t, \vec{p}) = \frac{2|Z_\pi|^2}{2E_\pi(\vec{p})} \left(e^{-E_\pi(\vec{p})t} + e^{-E_\pi(\vec{p})(2T-t)} + \dots \right) \quad (55)$$

cancelling the subleading exponential. This method requires two inversions and the calculation of

$$S_{A\pm P}(x, y) = S_A(x, y) \pm S_P(x, y) \quad (56)$$

Where the subscript refers to (A)ntiperiodic/(P)eriodic boundary conditions. Then,

$$C_\pm(t - \tau, 0) = - \sum_{\vec{x}\vec{y}} \text{Tr} [\gamma_5 \Gamma S_{A\pm P}(x, y) \bar{\Gamma}' \gamma_5 S_{A\pm P}(x, y)] \quad (57)$$

where $C_+(t - \tau, 0)$ gives the forward propagating part from 0 to T and $C_-(t - \tau, 0)$ gives the backwards propagating part from $2T$ to T .

6. Form Factors and Sequential Sources

The electromagnetic form factor of a 'pion' requires the evaluation of the matrix element

$$\langle \pi(p_f) | V_\mu | \pi(p_i) \rangle = (p_i + p_f)_\mu f(q^2) \quad (58)$$

where $q^2 = (p_i - p_f)^2$ and

$$V_\mu = q_u \bar{u} \gamma_\mu u + q_d \bar{d} \gamma_\mu d \quad (59)$$

is the electromagnetic current and q_i is the charge of the fermion i . This is the local (not conserved) current, so there will be a factor Z_V for renormalization. The matrix elements required look like:

$$C_3(t_f, t, t_i, \vec{p}_i, \vec{p}_f) = Z_V \sum_{\vec{x}\vec{y}\vec{z}} e^{-i\vec{p}_f(\vec{x}-\vec{y})} e^{i\vec{p}_i(\vec{y}-\vec{z})} \langle 0 | \bar{u} \gamma_5 d(\vec{x}, t_f) V_0(\vec{y}, t) \bar{d} \gamma_5 u(\vec{z}, t_i) | 0 \rangle \quad (60)$$

We take the $\mu = 0$ component since this is statistically cleaner and also nonzero independent of the momentum direction. The contractions give three propagators eg. taking the $\bar{d} \gamma_\mu d$ part of V_μ ,

$$Z_V \sum_{\vec{x}\vec{y}\vec{z}} e^{-i\vec{p}_f(\vec{x}-\vec{y})} e^{-i\vec{p}_i(\vec{y}-\vec{z})} \text{Tr} [S_u(\vec{z}, t_i; \vec{x}, t_f) \gamma_5 S_d(\vec{x}, t_f; \vec{y}, t) \gamma_0 S_d(\vec{y}, t; \vec{z}, t_i) \gamma_5] \quad (61)$$

There are also disconnected contributions from contracting the two fermions in the current together but we ignore those. The usual sequential source trick consists of solving

$$S_u(\vec{x}, t; \vec{z}, t_i) = \sum_{\vec{y}, \tau} D_u(\vec{x}, t; \vec{y}, \tau) \delta(\vec{y}, \tau; \vec{z}, t_i) \quad (62)$$

to get the point-to-all propagator (for a specific $\vec{z}$ and t_i as well as dropping the sum over $\vec{z}$ and using translational invariance). Then taking a single timeslice of the propagator $S_u(\vec{x}, t_f; \vec{z}, t_i)$ and solving,

$$D_d(\vec{x}, t_f; \vec{y}, t) G_{du}(\vec{y}, t; \vec{p}_f; t_f; \vec{z}, t_i) = e^{i\vec{p}_f \vec{x}} \gamma_5 S_u(\vec{x}, t_f; \vec{z}, t_i) \quad (63)$$

$$G_{du}(\vec{y}, t; \vec{p}_f; t_f; \vec{z}, t_i) = \sum_{\vec{x}} e^{i\vec{p}_f \vec{x}} S_d(\vec{y}, t; \vec{x}, t_f) \gamma_5 S_u(\vec{x}, t_f; \vec{z}, t_i) \quad (64)$$

to get the all-to-all-to-point contribution. Then

$$\gamma_5 [G_{du}(\vec{y}, t; \vec{p}_f; t_f; \vec{z}, t_i)]^\dagger \gamma_5 = \sum_{\vec{x}} e^{-i\vec{p}_f \vec{x}} \gamma_5 S_u^\dagger(\vec{x}, t_f; \vec{y}, t) \gamma_5 \gamma_5 \gamma_5 \gamma_5 S_d^\dagger(\vec{z}, t_i; \vec{x}, t_f) \gamma_5 \quad (65)$$

$$= \sum_{\vec{x}} e^{-i\vec{p}_f \vec{x}} S_u(\vec{z}, t_i; \vec{x}, t_f) \gamma_5 S_d(\vec{x}, t_f; \vec{y}, t) \quad (66)$$

$$= G_{ud}(\vec{z}, t_i; t_f; \vec{p}_f; \vec{y}, t) \quad (67)$$

$$C_3(t_f, t, t_i, \vec{p}_i, \vec{p}_f) = Z_V \sum_{\vec{y}} e^{-i(\vec{p}_i - \vec{p}_f) \vec{y}} Tr [G_{ud}(\vec{z}, t_i; t_f; \vec{p}_f; \vec{y}, t) \gamma_0 S_d(\vec{y}, t; \vec{z}, t_i) \gamma_5] \quad (68)$$

This Z_V factor is unknown. We show how to calculate it later, or cancel it, but an alternative is to use the conserved vector current in place of the local current:

$$V_\mu = \frac{1}{2} [\bar{\psi}(x + \mu)(1 + \gamma_\mu) U_\mu^\dagger(x) \psi(x) - \bar{\psi}(x)(1 - \gamma_\mu) U_\mu^\dagger(x) \psi(x + \mu)] \quad (69)$$

The trace in ?? becomes

$$Tr[S_d(\vec{y}, t + 1; \vec{z}, t_i) \gamma_5 (1 + \gamma_0) U_0^\dagger(\vec{y}, t) G_{ud}(\vec{z}, t_i; t_f; \vec{p}_f; \vec{y}, t) - S_d(\vec{y}, t; \vec{z}, t_i) \gamma_5 (1 - \gamma_0) U_0(\vec{y}, t) G_{ud}(\vec{z}, t_i; t_f; \vec{p}_f; \vec{y}, t + 1)] \quad (70)$$

If we use this then all the following formulas are the same except $Z_V \rightarrow 1$.

There is an alternative that doesn't require the sequential source trick. Using the properties of our noise sources,

$$S_d(\vec{x}, t_f; \vec{y}, t) \approx \frac{1}{K} \sum_{i=0}^K |\psi^{(i)}(\vec{x}, t_f)\rangle \langle \eta^{(i)}(\vec{y}, t)| \quad (71)$$

the three point correlation function becomes,

$$Z_V \frac{1}{K} \sum_{i=0}^K \sum_{\vec{x}, \vec{y}} e^{-i\vec{p}_f(\vec{x} - \vec{y})} e^{i\vec{p}_i \vec{y}} Tr \left[\langle \eta^{(i)}(\vec{y}, t) | \gamma_0 S_d(\vec{y}, t; \vec{0}, 0) \gamma_5 S_u(\vec{0}, 0; \vec{x}, t_f) \gamma_5 | \psi^{(i)}(\vec{x}, t_f) \rangle \right] \quad (72)$$

$$Z_V \frac{1}{K} \sum_{i=0}^K \sum_{\vec{x}, \vec{y}} e^{-i\vec{p}_f(\vec{x} - \vec{y})} e^{i\vec{p}_i \vec{y}} Tr \left[\langle \eta^{(i)}(\vec{y}, t) | \gamma_0 S_d(\vec{y}, t; \vec{0}, 0) S_u^\dagger(\vec{x}, t_f; \vec{0}, 0) | \psi^{(i)}(\vec{x}, t_f) \rangle \right] \quad (73)$$

Using this method we can inject arbitrary momentum at the source without the need for extra inversions.

6.1. Two Point Function

A complete set of hadrons is given by,

$$\sum_n \frac{|n\rangle\langle n|}{2E_n V} \quad (74)$$

the first term is the pion. The two point function (from point sources) is,

$$C(t, \vec{p}) = \sum_{\vec{x}} e^{-i\vec{p}\vec{x}} \langle O_\pi(\vec{x}, t) O_\pi^\dagger(\vec{0}, 0) \rangle \quad (75)$$

$$= \sum_{\vec{x}} \sum_n e^{i\vec{p}\vec{x}} \frac{\langle 0 | O_\pi(\vec{x}, t) | n \rangle \langle n | O_\pi^\dagger(\vec{0}, 0) | 0 \rangle}{2E_n} \quad (76)$$

Use $\sum_{\vec{y}} e^{-i\vec{p}\vec{y}} O_n^\dagger(\vec{y}, 0) | 0 \rangle = | n(\vec{p}) \rangle$, the time evolution operator e^{-Ht} and also the fact that the lightest meson dominates the sum to get,

$$C(t, \vec{p}) = \sum_{\vec{x}\vec{y}\vec{z}} \sum_{\vec{p}'} e^{-i\vec{p}\vec{x}} \frac{\langle 0 | O_\pi(\vec{x}, 0) O_\pi^\dagger(\vec{y}, 0) e^{i\vec{p}'\vec{y}} | 0 \rangle \langle 0 | e^{i\vec{p}'\vec{z}} O_\pi(\vec{z}, 0) O_\pi^\dagger(\vec{0}, 0) | 0 \rangle}{2E_\pi(\vec{p}')} e^{-E_\pi(\vec{p}')t} \quad (77)$$

The sum over $\vec{p}'$ gives a delta function leaving,

$$C(t, \vec{p}) = \sum_{\vec{x}\vec{y}} e^{-i\vec{p}\vec{x}} \frac{\langle 0 | O_\pi(\vec{x}, 0) O_\pi^\dagger(\vec{y}, 0) | 0 \rangle \langle 0 | O_\pi(\vec{y}, 0) O_\pi^\dagger(\vec{0}, 0) | 0 \rangle}{2E_\pi(\vec{p})} e^{-E_\pi(\vec{p})t} \quad (78)$$

Translational invariance lets us write,

$$C(t, \vec{p}) = \sum_{\vec{x}\vec{y}} e^{-i\vec{p}(\vec{x}-\vec{y})} e^{-i\vec{p}\vec{y}} \frac{\langle 0 | O_\pi(\vec{0}, 0) O_\pi^\dagger(\vec{x}-\vec{y}, 0) | 0 \rangle \langle 0 | O_\pi(\vec{y}, 0) O_\pi^\dagger(\vec{0}, 0) | 0 \rangle}{2E_\pi(\vec{p})} e^{-E_\pi(\vec{p})t} \quad (79)$$

Now changing variables gives us two Fourier transforms,

$$C(t, \vec{p}) = \frac{\langle 0 | O_\pi(\vec{0}, 0) | \pi(p) \rangle \langle \pi(p) | O_\pi^\dagger(\vec{0}, 0) | 0 \rangle}{2E_\pi(\vec{p})} e^{-E_\pi(\vec{p})t} \quad (80)$$

and finally using the time evolution operator we get,

$$C(t, \vec{p}) = \frac{|Z_\pi|^2}{2E_\pi(\vec{p})} e^{-E_\pi(\vec{p})t} \quad (81)$$

where

$$Z_\pi = \langle \pi(p) | O_\pi^\dagger(\vec{0}, 0) | 0 \rangle \quad (82)$$

6.2. Three Point Function

In less detail we insert two complete sets of states into the correlator (point sources so $(\vec{x}_i, t_i) = (\vec{0}, 0)$)

$$\langle \pi(p_f) | V_\mu | \pi(p_i) \rangle = \langle 0 | O(\vec{p}_f, t_f) V_\mu(\vec{x}, t) O^\dagger(\vec{p}_i, t_i) | 0 \rangle \quad (83)$$

$$= \langle 0 | O(\vec{0}, 0) | \pi(\vec{p}_f) \rangle \frac{e^{-(t_f-t)E_\pi(\vec{p}_f)}}{2E_\pi(\vec{p}_f)} \langle \pi(\vec{p}_f) | V_\mu(\vec{0}, 0) | \pi(\vec{p}_i) \rangle \quad (84)$$

$$\times \frac{e^{-(t-t_i)E_\pi(\vec{p}_i)}}{2E_\pi(\vec{p}_i)} \langle \pi(\vec{p}_i) | O^\dagger(\vec{0}, 0) | 0 \rangle \quad (85)$$

$$= \frac{Z_{\pi,f}^\dagger Z_{\pi,i}}{4E(\vec{p}_f)E(\vec{p}_i)} \langle \pi(\vec{p}_f) | V_\mu(\vec{0}, 0) | \pi(\vec{p}_i) \rangle e^{-(t_f-t)E_\pi(\vec{p}_f)-(t-t_i)E_\pi(\vec{p}_i)}$$

if $t < t_f$ we have the backwards contribution and the exponential changes to

$$- e^{-(t-t_f)E_\pi(\vec{p}_f)-(T-t+t_i)E_\pi(\vec{p}_i)} \quad (86)$$

6.3. Correlator Ratios: Z_V

Z_V can be obtained as follows: The ratio,

$$\frac{C_{\rightarrow}(t_f, \vec{0})}{C_3(t_f, t, \vec{p}_i, \vec{p}_f)} = \frac{\frac{|Z_{\pi}(\vec{0})|^2}{2m_{\pi}} e^{-m_{\pi} t_f}}{\frac{|Z_{\pi}(\vec{0})|^2}{4m_{\pi}^2} \langle \pi(\vec{0}) | V_{\mu} | \pi(\vec{0}) \rangle e^{-(t_f-t)m_{\pi}-tm_{\pi}}} \quad (87)$$

$$= \frac{1}{\frac{1}{2m_{\pi}} \langle \pi(\vec{0}) | V_{\mu} | \pi(\vec{0}) \rangle} \quad (88)$$

$$= \frac{1}{\frac{1}{2m_{\pi}} 2m_p i f(0) / Z_V} = Z_V. \quad (89)$$

Where we used that the renormalized form factor $f(0) = 1$

6.4. Correlator Ratios: $f(q)$

There are various ways to cancel the unwanted terms and get $f(q)$,

6.4.1. RBC-UKQCD Ratio

We examine the ratio,

$$2m_{\pi} \frac{C_3(t, t_f, \vec{p}, \vec{0}) C_{\rightarrow}(t, \vec{0})}{C_3(t, t_f, \vec{0}, \vec{0}) C_{\rightarrow}(t, \vec{p})} \quad (90)$$

Assuming Z_{π} is momentum independant (this also works (probably) if $Z_{\pi} = E(\vec{p}) f_{\pi}$, which is the case for $O = \bar{u} \gamma_0 \gamma_5 d$ type interpolators) the numerator is,

$$\frac{Z_V |Z_{\pi}|^2}{4E(\vec{p})E(\vec{0})} f(q^2)(E(\vec{p}) + m_{\pi}) \frac{|Z_{\pi}|^2}{2E(\vec{0})} e^{-E(\vec{p})t - E(\vec{0})(t_f-t) - E(\vec{0})t} \quad (91)$$

and the denominator is,

$$\frac{Z_V |Z_{\pi}|^2}{4E(\vec{0})E(\vec{0})} f(0)(m_{\pi} + m_{\pi}) \frac{|Z_{\pi}|^2}{2E(\vec{p})} e^{-E(\vec{0})t - E(\vec{0})(t_f-t) - E(\vec{p})t} \quad (92)$$

Cancelling leaves,

$$2m_{\pi} \frac{C_3(t, t_f, \vec{p}, \vec{0}) C_{\rightarrow}(t, \vec{0})}{C_3(t, t_f, \vec{0}, \vec{0}) C_{\rightarrow}(t, \vec{p})} = f(q^2)(E(\vec{p}) + m_{\pi}) \quad (93)$$

note there is no Z_V here.

6.4.2. Bonnet et. al. Ratio

$$\frac{2Z_V m_{\pi}}{E(\vec{p}) + m_{\pi}} \frac{C_3(t, t_f, \vec{p}, \vec{0}) C_{\rightarrow}(t, \vec{0})}{C_{\rightarrow}(t, \vec{p}) C_{\rightarrow}(t_f, \vec{0})} \quad (94)$$

the numerator of the right term is,

$$\frac{|Z_{\pi}|^2}{4E(\vec{p})m_{\pi}} f_B(q^2)(E(\vec{p}) + m_{\pi}) \frac{|Z_{\pi}|^2}{2m_{\pi}} e^{-E(\vec{p})t - m_{\pi}(t_f-t) - m_{\pi}t} \quad (95)$$

the denominator of the right term is,

$$\frac{|Z_{\pi}|^2}{2m_{\pi}} \frac{|Z_{\pi}|^2}{2E(\vec{p})} e^{-m_{\pi}t - m_{\pi}(t_f-t) - E(\vec{p})t} \quad (96)$$

Cancelling leaves,

$$\frac{f_B(q^2)(E(\vec{p}) + m_{\pi})}{2E(\vec{p})} \quad (97)$$

the kinematic factors are cancelled

$$\frac{2Z_V m_{\pi}}{E(\vec{p}) + m_{\pi}} \frac{f_B(q^2)(E(\vec{p}) + m_{\pi})}{2E(\vec{p})} = Z_V f_B(q^2) = f(q^2) \quad (98)$$

you need to actually know Z_V or use the conserved current.